

Day 2 - Facilitation Guide

(Working with HTML)

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(2 hrs) ILT

I. Recap of the previous session

In the last session we gave you an introduction to HTML. HTML (HyperText Markup Language) is considered as the standard markup language for creating Web pages. You have learned the importance of HTML in Web development in the session.

We learnt the basic structure of an html file and also gave you an overview of basic HTML tags. We gave an example of a simple HTML file. You also learned how to create an HTML file and open it in a browser. We illustrated with examples how HTML is extensively used in real life applications.

In this session we will deepen our knowledge on HTML. We will explain some of the important tags used in a HTML document. We will also discuss in detail about Text formatting tags, List Tags and Image Tags.

II. HTML Tags and Elements

HTML (HyperText Markup Language) uses tags to define elements within a document. Tags are the building blocks of HTML and are enclosed in angle brackets < >. Tags usually come in pairs: an opening tag and a closing tag. The opening tag signifies the beginning of an element, and the closing tag signifies the end of that element. Tags are **not case-sensitive**, meaning <tag> is the same as <Tag> or <TAG>.

Example of an opening and closing tag pair:

`<p>This is a paragraph.</p>`

In this example:

- `<p>` is the opening tag.
- `</p>` is the closing tag.
- "This is a paragraph." is the content of the `<p>` element.

Some HTML tags, like line breaks (`<br>`) and images (`<img>`), don't have a closing tag because they don't have any content between an opening and closing tag. They are self-closing tags.

HTML Elements:

An element is a complete set of tags and the content between them. Elements can be as simple as a single word or as complex as multiple paragraphs with images, links, lists, etc.

Here's an example of a more complex HTML element:

`<a href="https://www.amazon.com"> Click here to visit Example Website </a>`

In this example:

- `<a>` is the opening tag for a hyperlink (anchor) element.
- `</a>` is the closing tag.
- "https://www.amazon.com" is an attribute of the `<a>` element specifying the destination URL.
- "Click here to visit Amazon Website" is the content of the `<a>` element, which is the visible part of the hyperlink.

Attributes provide additional information about an element and are always included in the opening tag. They usually come in name/value pairs like `name="value"`. For example, in the `<a>` tag above, `href` is the attribute name, and "https://www.amazon.com" is the attribute value.

HTML elements can be nested inside one another, creating a hierarchical structure for the content of a Web page. Understanding how to use HTML tags and elements is fundamental to Web development as they form the backbone of Web pages, allowing developers to create structured and meaningful content.

III. Text formatting tags

In HTML, text formatting tags are used to change the appearance or style of text within a Web page. They allow you to control how the text is displayed, such as making it bold, italic, underlined, or specifying its size and color. Here are some commonly used text formatting tags in HTML:

Bold Text:

`<b>`: Renders text in bold.

`<strong>`: Represents strong importance, typically displayed as bold.

Example:

`<b>This is bold text.</b>`

`<strong>This is also bold text.</strong>`

Italic Text:

`<i>`: Renders text in italic.

`<em>`: Represents emphasized text, typically displayed as italic.

Example:

`<i>This is italic text.</i>`

`<em>This is emphasized text.</em>`

Underlined Text:

`<u>`: Renders text with an underline.

Example:

`<u>This text is underlined.</u>`

Strikethrough Text:

`<s>` or `<strike>`: Renders text with a strikethrough line.

Example:

`<s>This text has a strikethrough.</s>`

`<strike>This text also has a strikethrough.</strike>`

Subscript and Superscript Text:

`<sub>`: Renders text as subscript (e.g., H`<sub>2</sub>`O).

`<sup>`: Renders text as superscript (e.g., x`<sup>2</sup>`).

Example:

H`<sub>2</sub>`O (Subscript)

x² (Superscript)

Changing Text Size:

<h1> to <h6>: Headings of decreasing sizes, with <h1> as the largest and <h6> as the smallest.

: Changes the text size, where n is a number (1 to 7) representing the size relative to the default size.

Example:

<h1>This is Heading 1</h1>

<h2>This is Heading 2</h2>

This text has a custom size.

Changing Text Color:

: Changes the text color, where color_name can be a color keyword (like "red") or a hexadecimal color code (#RRGGBB).

Example:

This text is blue.

This text is red.

Note: It's important to note that some of these tags, especially the tag, are considered deprecated in modern HTML standards. Instead, CSS (Cascading Style Sheets) is the preferred method for controlling the visual presentation of Web content. However, understanding these tags can still be useful when working with older HTML code or when learning the basics of Web development

HTML examples showing all of the above tags:

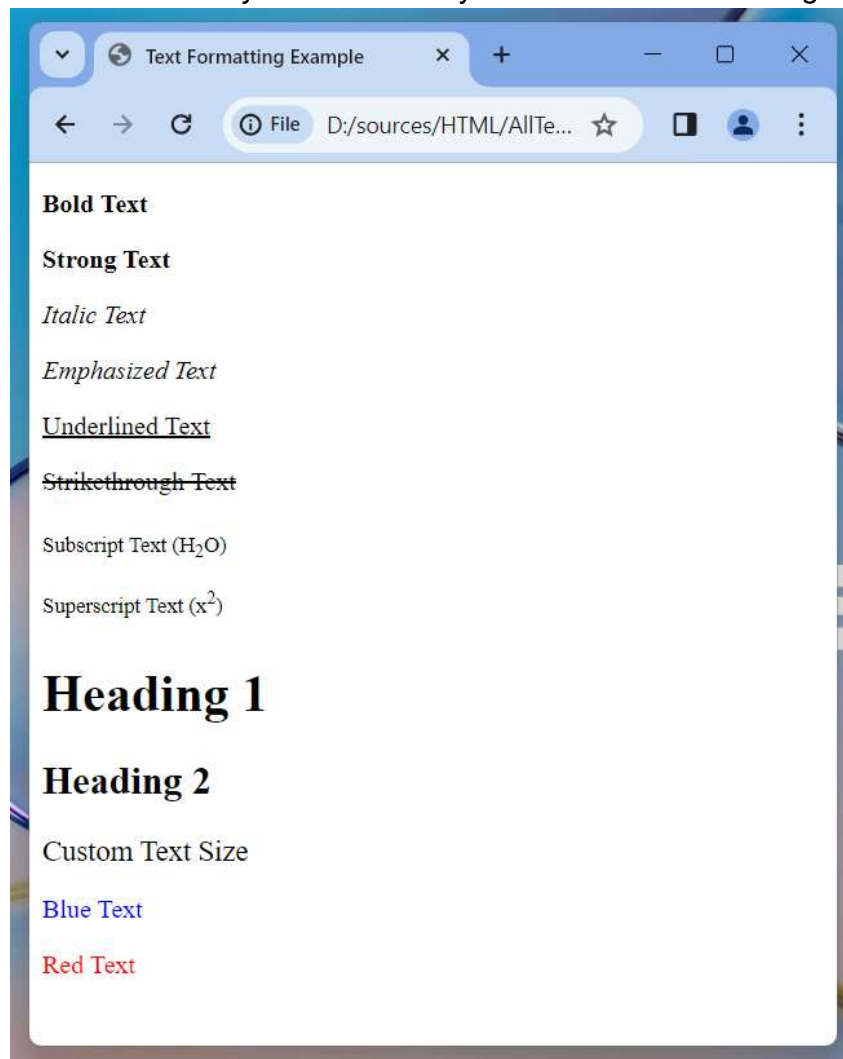
```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Text Formatting Example</title>
  </head>
  <body>
    <p><b>Bold Text</b></p>
    <p><strong>Strong Text</strong></p>
    <p><i>Italic Text</i></p>
    <p><em>Emphasized Text</em></p>
    <p><u>Underlined Text</u></p>
    <p><s>Strikethrough Text</s></p>
```

```

<p><sub>Subscript Text (H<sub>2</sub>O)</sub></p>
<p><sup>Superscript Text (x<sup>2</sup>)</sup></p>
<p><h1>Heading 1</h1></p>
<p><h2>Heading 2</h2></p>
<p><font size="4">Custom Text Size</font></p>
<p><font color="blue">Blue Text</font></p>
<p><font color="#FF0000">Red Text</font></p>
</body>
</html>

```

If you run the above code in any Web browser you can see the following output:



IV. Style tag

In HTML, the <style> tag is used to embed or define styles directly within an HTML document. The styles defined inside the <style> tag apply to the elements within that document. The <style> tag is typically placed within the <head> section of an HTML document.

HTML styling can be done using style tag and style attributes. Let's explore both the <style> tag and style attributes:

Syntax:

<tagname style="property:value;">

The tagname includes <p>, <body>, from <h1> to <h6> etc.

The property is borrowed from CSS like color, font-size, font-family etc.

The value is also borrowed from CSS.

Style Attributes

Style attributes are added directly to HTML elements using the style attribute. This inline styling approach allows you to apply styles directly to specific elements without using an external or internal stylesheet.

- The <style> tag is opened with <style> and closed with </style>.
- The rules (styles) are written between the opening and closing tags.
- Each rule consists of a name and a declaration block within parenthesis. The name is called selector, selector targets HTML elements, and the declaration block contains the style properties and values.

List of attributes:

Listed below are some commonly used style attributes:

HTML Font Family

The font family changes the font style of a text and can be used in any text writing tag like <p> or heading tag. These font families include all the names that you find in Microsoft Office or any other writing-based software

Eg:

```
<h1 style="font-family:commanders;">
    Hello World
</h1>
<h2 style="font-family:Chaparral Pro Light;">
    Hello World
</h2>
<h3 style="font-family:algerian;">
    Hello World
</h3>
<p style="font-family:Castellar;">
    Hello World
```

</p>

Output:

Hello World

Hello World

HELLO WORLD

HELLO WORLD

HTML Font Size

The font size changes the size of a text and this can also be used in any text writing tag like <p> or heading tag. The units can be given in “%” or pixels or other units can also be included.

Eg:

```
<h1 style="font-size:80%;">
  Hello World
</h1>
<h2 style="font-size:150%;">
  Hello World
</h2>
<h3 style="font-size:20px;">
  Hello World
</h3>
<p style="font-size:30px;">
  Hello World
</p>
```

Output:

Hello World

Hello World

Hello World

Hello World

HTML Font Color

The font color tag changes the color of a text and can be used in any text writing tag like <p> or heading tag. We can use both names of the colors or also the color codes that are mainly used in Photoshop.

For various color codes or to pick from various color ranges refer HTML Color Codes.
<https://htmlcolorcodes.com/>

Eg:

```
<h1 style="color:red;">
    Hello World
</h1>
<h2 style="color:#8CCEF9;">
    Hello World
</h2>
<h3 style="color:green;">
    Hello World
</h3>
<p style="color:#810CA6;">
    Hello World
</p>
```

Output:

Hello World

Hello World

Hello World

Hello World

HTML Text Align

The text alignment tag is used to change the alignment of a text including centre, left or right alignment.

Eg:

```
<h1 style="text-align:left;">
    Hello World
</h1>
<h2 style="text-align:center;">
    Hello World
</h2>
<p style="text-align:right;">
    Hello World
</h2>
```

Output:

Hello World

Hello World

Hello World

HTML Background Color

Using this attribute we can change the color of the background page or Web page. This attribute is used along with the body tag to change the whole color of the body. It can also be used along with the text tags to change the text block's color.

Eg:

```
<h1 style="font-family:commanders;
    background-color:yellow;">
    Hello World
</h1>
<h2 style="font-family:algerian;
    background-color:cyan;">
    Hello World
</h2>
<p style="font-family:Castellar;
    background-color:green;">
```

```
Hello World  
</p>
```

Output:

Hello World

HELLO WORLD

HELLO WORLD

Selectors

In HTML's <style> tag, **selectors** are used to target and apply styles to specific HTML elements.

Here are some common types of selectors used in style tag:

Universal Selector (*):

Selects all elements on the page.

```
* {  
  margin: 0;  
  padding: 0;  
}
```

Type or Element Selector:

Selects all instances of a specific HTML element.

```
h1 {  
  color: #333;  
}  
  
p {  
  font-size: 16px;  
}
```

In the above example, h1 and p are selectors.

These selectors provide a powerful way to target and style specific elements or groups of elements within an HTML document. They allow developers to apply styles based on various criteria, making the styling process flexible and dynamic.

Here's an example on how the <style> tag is used with selectors in an HTML page:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTML Document with Styles</title>

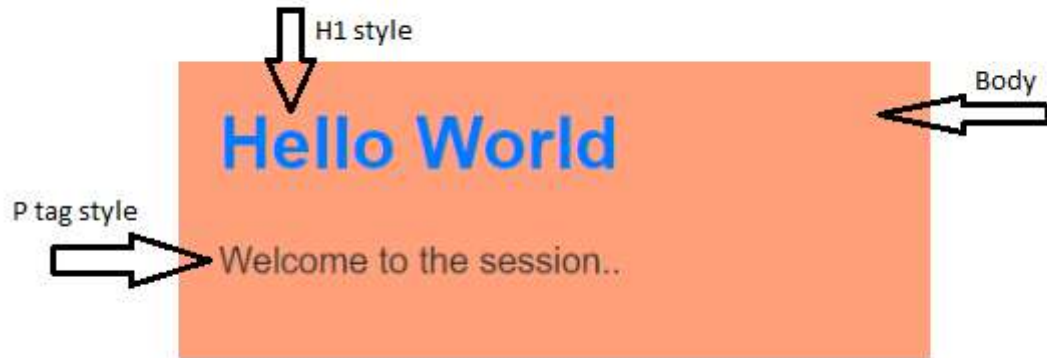
  <!-- The <style> tag is used to define styles -->
  <style>
    /* the style rule for body tag */
    body {
      font-family: Arial, sans-serif;
      background-color: #FFA07A;
      color: #333;
      margin: 20px;
    }
    /* style rule for h1 tag */
    h1 {
      color: #007BFF;
    }

    /* style rule for paragraph tag */
    p {
      line-height: 1.5;
    }

    /* Add more styles as needed */
  </style>
</head>
<body>

  <h1>Hello World</h1>
  <p>Welcome to the session..</p>
</body>
</html>
```

Output:



In the example above:

- The **body** selector targets the entire document body and sets properties such as font-family, background-color, and margin.
- The **h1** selector targets all `<h1>` elements and sets the text color to #007BFF (a shade of blue).
- The **p** selector targets all `<p>` elements and sets the line-height property.

In summary, the `<style>` tag is a way to include styles directly within an HTML document. It's useful for small projects or when you want to encapsulate styles within a specific document. However, External CSS (Cascading Style Sheets) files provide a more organized and maintainable way to manage styles, especially for larger projects. External stylesheets will be discussed in more detail in the next session.

V. Lists

Lists in HTML are used to organize and present information in a structured manner. There are three types of lists in HTML: unordered lists, ordered lists, and definition lists. Let me explain each type with examples:

1. Unordered Lists (`<ul>`)

Unordered lists are used to create a list of items where the order doesn't matter. Each item in an unordered list is preceded by a bullet point.

Example:

```
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ul>
```

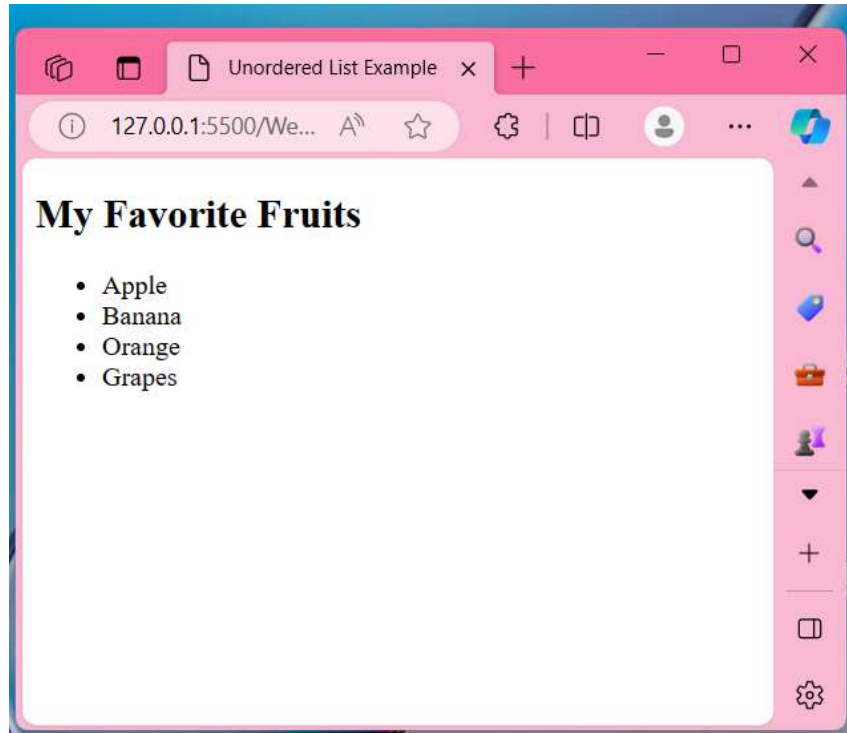
In this example, you'll get an unordered list like this:

- Item 1
- Item 2
- Item 3

Here is an example of HTML code to create an unordered list.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Unordered List Example</title>
</head>
<body>
  <h2>My Favorite Fruits</h2>
  <ul>
    <li>Apple</li>
    <li>Banana</li>
    <li>Orange</li>
    <li>Grapes</li>
  </ul>
</body>
</html>
```

If you open the above HTML code in Web browser then you will see:



2. Ordered Lists ()

Ordered lists are used when the order of the items is important. Each item in an ordered list is preceded by a number (or another marker based on the list style type).

Example:

```
<ol>
  <li>First item</li>
  <li>Second item</li>
  <li>Third item</li>
</ol>
```

In this example, you'll get an ordered list like this:

```
First item
Second item
Third item
```

Let us create an HTML page with an ordered list .

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Grocery List</title>
```

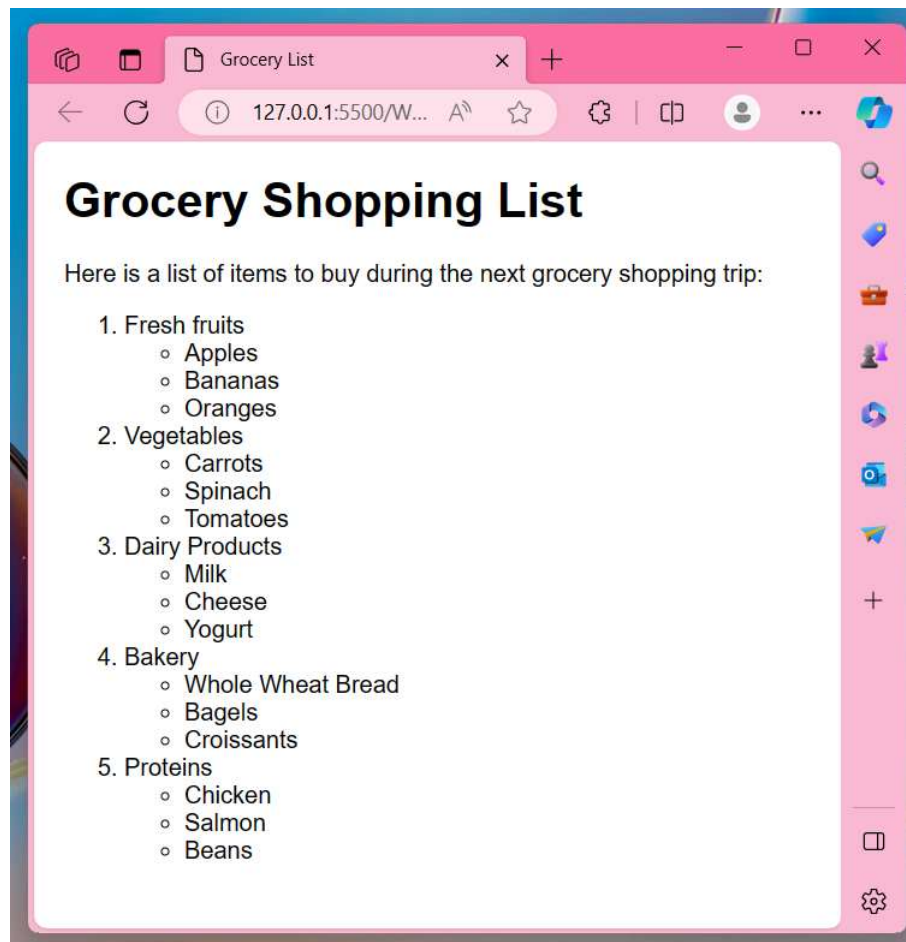
```
<style>
  body {
    font-family: Arial, sans-serif;
    margin: 20px;
  }
</style>
</head>
<body>

  <h1>Grocery Shopping List</h1>
  <p>Here is a list of items to buy during the next grocery shopping trip:</p>

  <ol>
    <li>Fresh fruits
      <ul>
        <li>Apples</li>
        <li>Bananas</li>
        <li>Oranges</li>
      </ul>
    </li>
    <li>Vegetables
      <ul>
        <li>Carrots</li>
        <li>Spinach</li>
        <li>Tomatoes</li>
      </ul>
    </li>
    <li>Dairy Products
      <ul>
        <li>Milk</li>
        <li>Cheese</li>
        <li>Yogurt</li>
      </ul>
    </li>
    <li>Bakery
      <ul>
        <li>Whole Wheat Bread</li>
        <li>Bagels</li>
        <li>Croissants</li>
      </ul>
    </li>
    <li>Proteins
      <ul>
        <li>Chicken</li>
        <li>Salmon</li>
        <li>Beans</li>
      </ul>
    </li>
  </ol>
```

```
</body>
</html>
```

If you copy this code in an HTML file, save it and open then you will find:



We can also style the ordered list using `<style>` tags. But keep in mind that if we use style tags then automatic numbering can be changed also. For an ordered list the value of `list-style-type` is `decimal`. If we do not want numbering then change it to `none`.

```
ol {
  ... Other attributes
  list-style-type: decimal; <!-- change to none for no numbering -->
  ...
}
```

Here is the above example with internal styling.

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
```



```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Grocery List</title>
<style>
  body {
    font-family: 'Arial', sans-serif;
    margin: 20px;
    background-color: #f9f9f9;
    color: #333;
  }

  h1 {
    color: #3498db;
    text-align: center;
  }

  p {
    font-size: 1.2em;
    margin-bottom: 20px;
    text-align: center;
  }

  ul {
    list-style-type: none;
    padding: 0;
    margin-top: 0;
  }

  li ul {
    color: #350303;
    font-weight: 500;
    margin-top: 5px;
    margin-bottom: 10px;
    margin-left: 60px;
    list-style-type: none;
  }

  ol {
    color: #350363;
    font-weight: bold;
    margin-left: 20px;
    font-size: 20px;
    list-style-type: decimal;
  }
}
```

```
</style>
</head>
```

```
<body>
```

```
<h1>Grocery Shopping List</h1>
```

```
<p>Here is a list of items to buy during the next grocery shopping trip:</p>
```

```
<ol>
```

```
<li>Fresh fruits
```

```
<ul>
```

```
<li>Apples</li>
```

```
<li>Bananas</li>
```

```
<li>Oranges</li>
```

```
</ul>
```

```
</li>
```

```
<li>Vegetables
```

```
<ul>
```

```
<li>Carrots</li>
```

```
<li>Spinach</li>
```

```
<li>Tomatoes</li>
```

```
</ul>
```

```
</li>
```

```
<li>Dairy Products
```

```
<ul>
```

```
<li>Milk</li>
```

```
<li>Cheese</li>
```

```
<li>Yogurt</li>
```

```
</ul>
```

```
</li>
```

```
<li>Bakery
```

```
<ul>
```

```
<li>Whole Wheat Bread</li>
```

```
<li>Bagels</li>
```

```
<li>Croissants</li>
```

```
</ul>
```

```
</li>
```

```
<li>Proteins
```

```
<ul>
```

```
<li>Chicken</li>
```

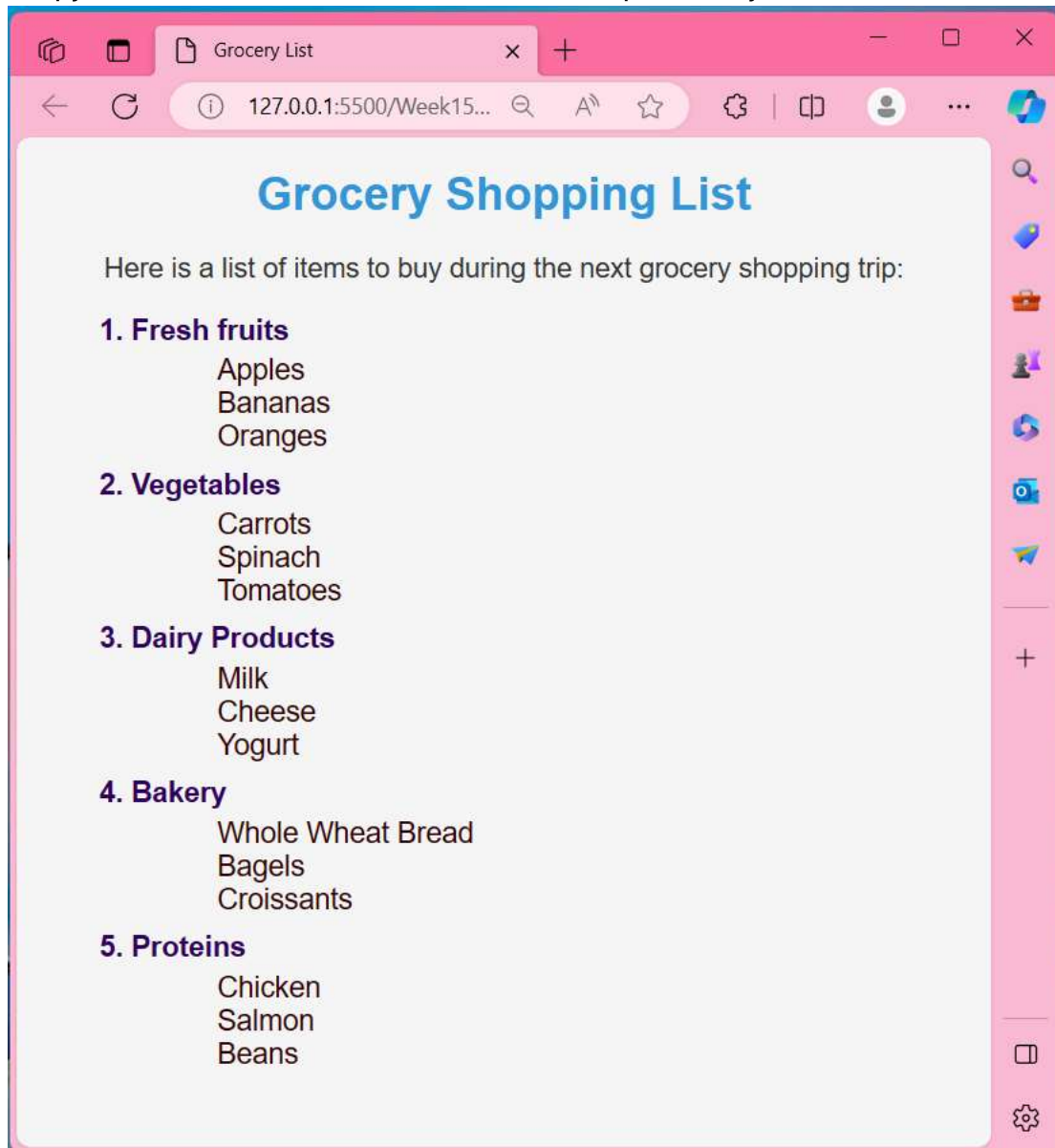
```
<li>Salmon</li>
```

```
<li>Beans</li>
```

```
</ul>
</li>
</ol>
</body>

</html>
```

If you copy this code in an HTML file, save it and open then you will find:



3. Definition Lists (<dl>)

Definition lists are used to create a list of terms and their corresponding definitions. Each term is enclosed in a <dt> (definition term) tag, and its definition is enclosed in a <dd> (definition description) tag.

Example:

```
<dl>
  <dt>Term 1</dt>
  <dd>Definition 1</dd>
  <dt>Term 2</dt>
  <dd>Definition 2</dd>
</dl>
```

In this example, you'll get a definition list like this:

Term 1
Definition 1
Term 2
Definition 2
Nested Lists

Below is an example of a definition list to list software courses and their brief content.

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Software Course Subjects</title>
    <style>
      body {
        font-family: Arial, sans-serif;
        margin: 20px;
      }

      dl {
        margin-bottom: 20px;
      }

      dt {
        font-weight: bold;
        margin-bottom: 5px;
      }

      dd {
        margin-left: 20px;
        margin-bottom: 10px;
      }
    </style>
  </head>
  <body>
```

```
<h1>Software Course Subjects</h1>

<dl>
  <dt>Introduction to Programming</dt>
  <dd>An overview of fundamental programming concepts, syntax, and
problem-solving techniques.</dd>
</dl>

<dl>
  <dt>Web Development Fundamentals</dt>
  <dd>Introduction to Web technologies, including HTML, CSS, and basic
JavaScript.</dd>
</dl>

<dl>
  <dt>Database Management</dt>
  <dd>Exploration of database concepts, SQL, and relational database
management systems (RDBMS).</dd>
</dl>

<dl>
  <dt>Object-Oriented Programming (OOP)</dt>
  <dd>Understanding the principles of OOP, including classes, objects,
inheritance, and polymorphism.</dd>
</dl>

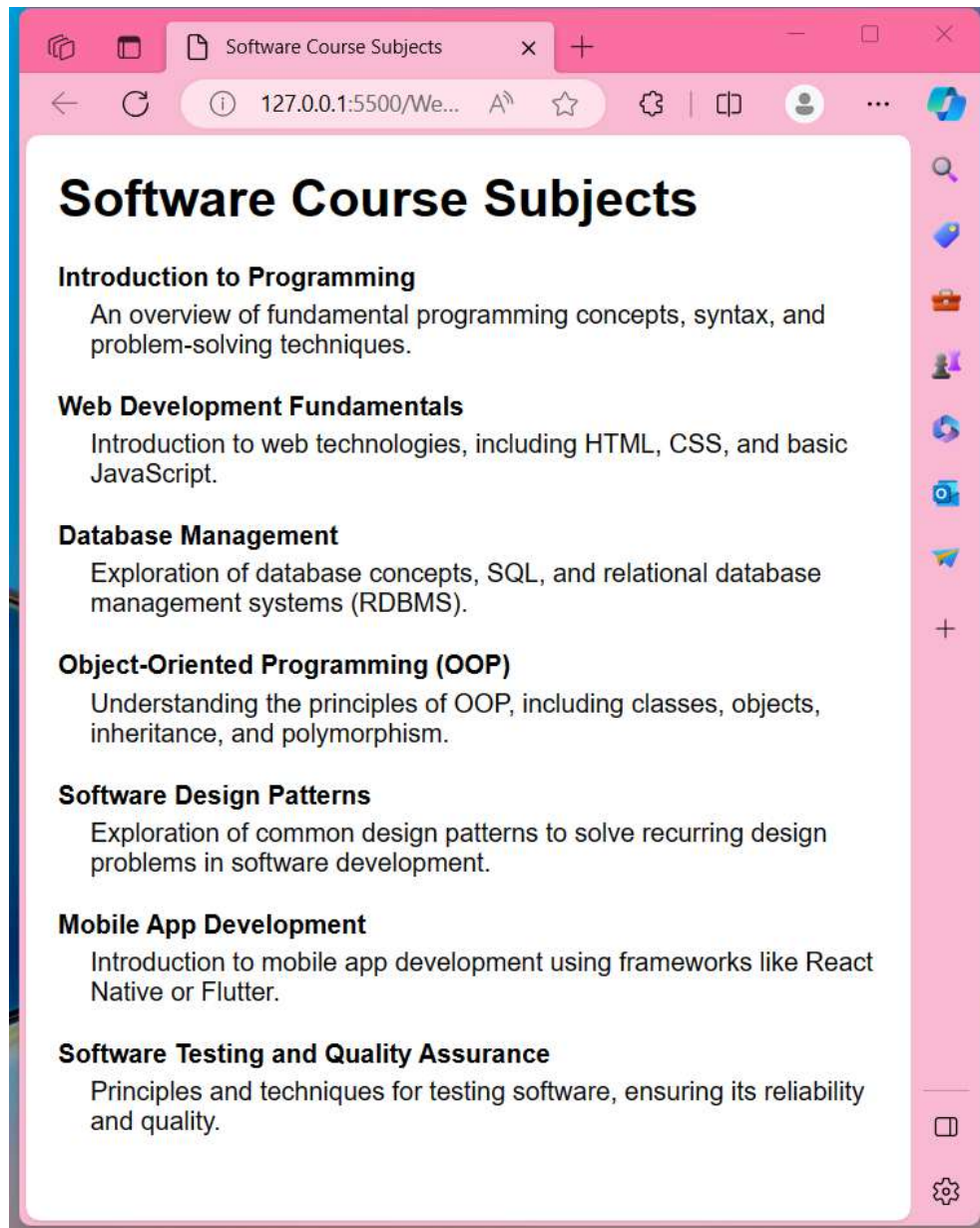
<dl>
  <dt>Software Design Patterns</dt>
  <dd>Exploration of common design patterns to solve recurring design
problems in software development.</dd>
</dl>

<dl>
  <dt>Mobile App Development</dt>
  <dd>Introduction to mobile app development using frameworks like React
Native or Flutter.</dd>
</dl>

<dl>
  <dt>Software Testing and Quality Assurance</dt>
  <dd>Principles and techniques for testing software, ensuring its reliability
and quality.</dd>
</dl>

</body>
</html>
```

If you copy this code in an HTML file, save it and open then you will find:



We can change the color scheme of the above HTML page by changing the color scheme in the style tag. Here is another example by changing the style of the above html page.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Software Course Subjects</title>
<style>
  body {
    font-family: Arial, sans-serif;
    background-color: #f0f0f0;
    margin: 0;
    padding: 0;
  }

  header {
    background-color: #fe5050;
    color: #fff;
    text-align: center;
    padding: 1px;
  }

  section {
    max-width: 800px;
    margin: 20px auto;
    padding: 20px;
    background-color: #cf7b7b;
    border-radius: 8px;
    box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
  }

  h1 {
    background-color: #1877F2;
    color: #fff;
    text-align: center;
    padding: 1px;
  }

  dd {
    line-height: 1.6;
    color: #555;
  }

  dt {
    color: #350303;
    font-weight: 700;
    font-size: 18px;
  }
</style>
</head>
```

<body>

<h1>Software Course Subjects</h1>

<dl>

<dt>Introduction to Programming</dt>

<dd>An overview of fundamental programming concepts, syntax, and problem-solving techniques.</dd>

</dl>

<dl>

<dt>Web Development Fundamentals</dt>

<dd>Introduction to Web technologies, including HTML, CSS, and basic JavaScript.</dd>

</dl>

<dl>

<dt>Database Management</dt>

<dd>Exploration of database concepts, SQL, and relational database management systems (RDBMS).</dd>

</dl>

<dl>

<dt>Object-Oriented Programming (OOP)</dt>

<dd>Understanding the principles of OOP, including classes, objects, inheritance, and polymorphism.</dd>

</dl>

<dl>

<dt>Software Design Patterns</dt>

<dd>Exploration of common design patterns to solve recurring design problems in software development.</dd>

</dl>

<dl>

<dt>Mobile App Development</dt>

<dd>Introduction to mobile app development using frameworks like React Native or Flutter.</dd>

</dl>

<dl>

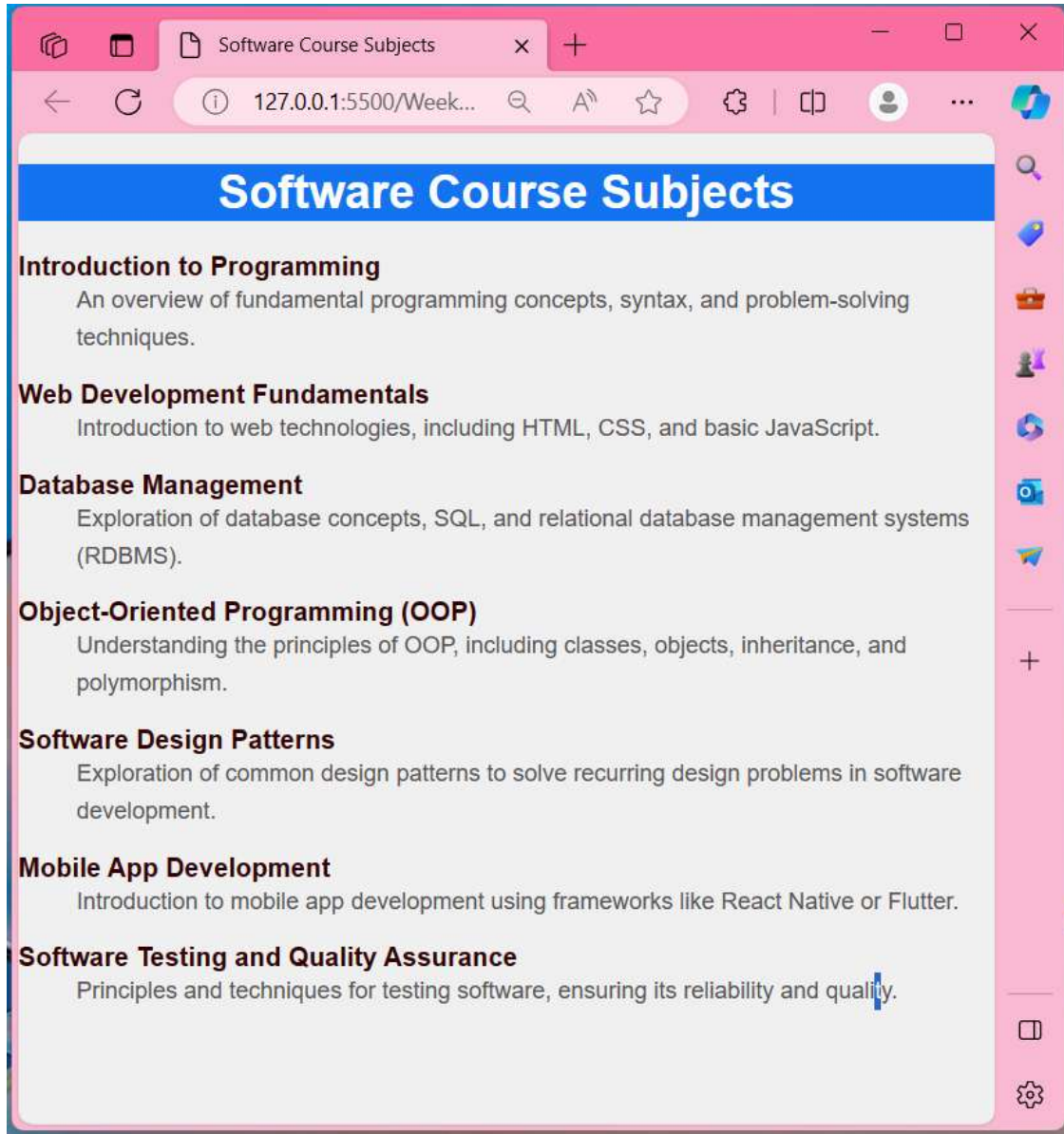
<dt>Software Testing and Quality Assurance</dt>

<dd>Principles and techniques for testing software, ensuring its reliability and quality.</dd>


```
</dl>

</body>
</html>
```

Now if you open the above HTML page in a browser you will see:



You can also nest lists inside one another to create a hierarchical structure.

Example:

```
<ul>
  <li>Item 1</li>
```

```

<li>Item 2
  <ul>
    <li>Subitem 1</li>
    <li>Subitem 2</li>
  </ul>
</li>
<li>Item 3</li>
</ul>

```

In this example, "Item 2" has a sublist with "Subitem 1" and "Subitem 2".

Here is a complete example of how a nested list can be created within an HTML page.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Nested Lists Example</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 20px;
    }
  </style>
</head>
<body>

  <h1>Menu</h1>

  <ul>
    <li>Appetizers
      <ul>
        <li>Spring Rolls</li>
        <li>Stuffed Mushrooms</li>
      </ul>
    </li>
    <li>Main Courses
      <ul>
        <li>Grilled Salmon
          <ul>
            <li>Salmon with Lemon Butter</li>
            <li>Teriyaki Glazed Salmon</li>
          </ul>
        </li>
      </ul>
    </li>
  </ul>

```

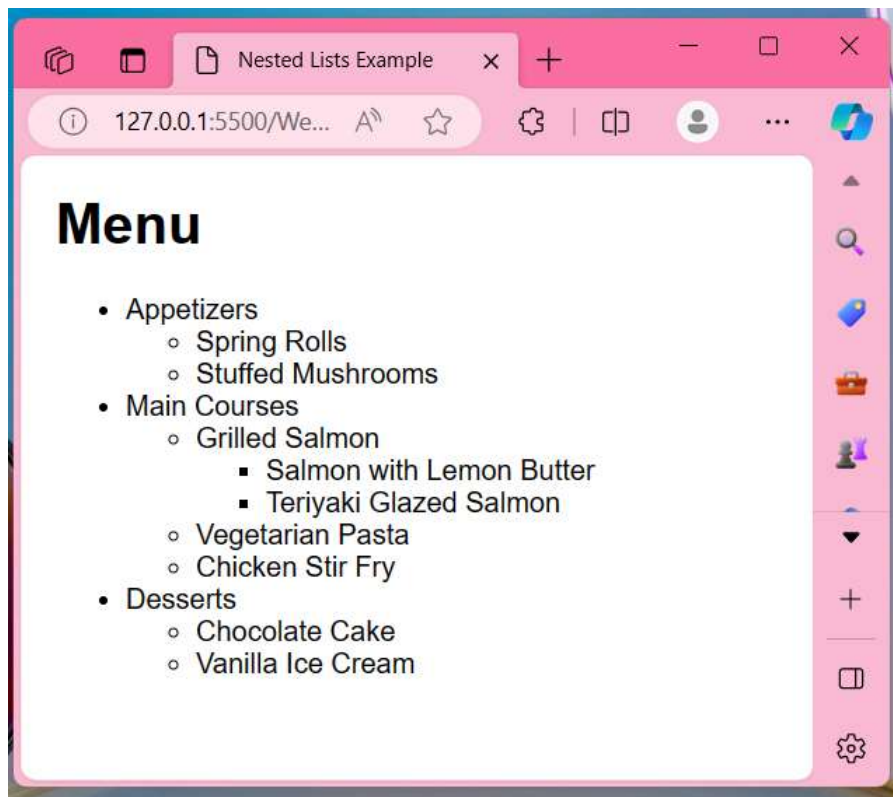
```

        </li>
        <li>Vegetarian Pasta</li>
        <li>Chicken Stir Fry</li>
    </ul>
</li>
<li>Desserts
    <ul>
        <li>Chocolate Cake</li>
        <li>Vanilla Ice Cream</li>
    </ul>
</li>
</ul>

</body>
</html>

```

If you copy this code in an HTML file, save it and open then you will find:



Lists are crucial for organizing and structuring content on Web pages, providing clarity and making the information easily readable for users.

VI. Links

HTML links, also known as hyperlinks, are used to connect one Web page to another or to link to various resources on the Web, such as documents, images, videos, or other types of files. Links are a fundamental part of the Web, allowing users to navigate between different pages and access diverse content.

Here's how you create links in HTML:

Basic Link Structure:

```
<a href="URL">Link Text</a>
```

<a>: Anchor tag, used to define a hyperlink.

- href="URL": Specifies the URL (Uniform Resource Locator) of the linked resource.
- Link Text: The visible text that users click on to navigate to the linked resource.

- **Examples:**

1. Linking to Another Web Page:

```
<a href="https://www.example.com">Visit Example Website</a>
```

In this example, the text "Visit Example Website" is a clickable link that directs users to the "https://www.example.com" Web page.

2. Linking to an Email Address:

```
<a href="mailto:example@example.com">Send Email</a>
```

By using mailto:, you can create a link that opens the user's default email client and populates the recipient email address.

3. Linking Within the Same Page (Using Anchors):

```
<a href="#section2">Go to Section 2</a>
```

```
<!-- ...some content... -->
```

```
<h2 id="section2">Section 2</h2>
```

```
<p>This is the content of Section 2.</p>
```

In this example, clicking "Go to Section 2" will scroll the page to the section with the ID "section2".

4. Linking to a File (e.g., PDF, Image, etc.):

```
<a href="document.pdf">Download PDF</a>
```

Here, "document.pdf" is the name of the file in the same directory as the HTML file. Clicking "Download PDF" will prompt the user to download the PDF file.

5. Opening a Link in a New Tab/Window:

```
<a href="https://www.example.com" target="_blank">Visit Example Website</a>
```

Adding target="_blank" to the <a> tag opens the link in a new tab or window, depending on the user's browser settings.

6. Linking to a Different Section on Another Page:

```
<a href="another-page.html#section3">Go to Section 3 on Another Page</a>
```

In this example, the link navigates to a specific section with the ID "section3" on the "another-page.html" page.

HTML links are essential for creating a seamless and interconnected Web experience, allowing users to explore various content and navigate between different parts of a Website or different Websites altogether.

Here's an example of all types of HTML links (hyperlinks) in a single HTML file:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTML Links Example</title>
</head>
<body>
  <h2>Basic Links:</h2>
  <ul>
    <li><a href="https://www.example.com">Visit Example Website</a></li>
    <li><a href="mailto:info@example.com">Send Email</a></li>
    <li><a href="#section2">Go to Section 2</a></li>
  </ul>

  <h2>Links with Target Attribute:</h2>
  <ul>
    <li><a href="https://www.example.com" target="_blank">Visit Example
Website (Opens in a new tab/window)</a></li>
  </ul>

  <h2>File Links:</h2>
  <ul>
```

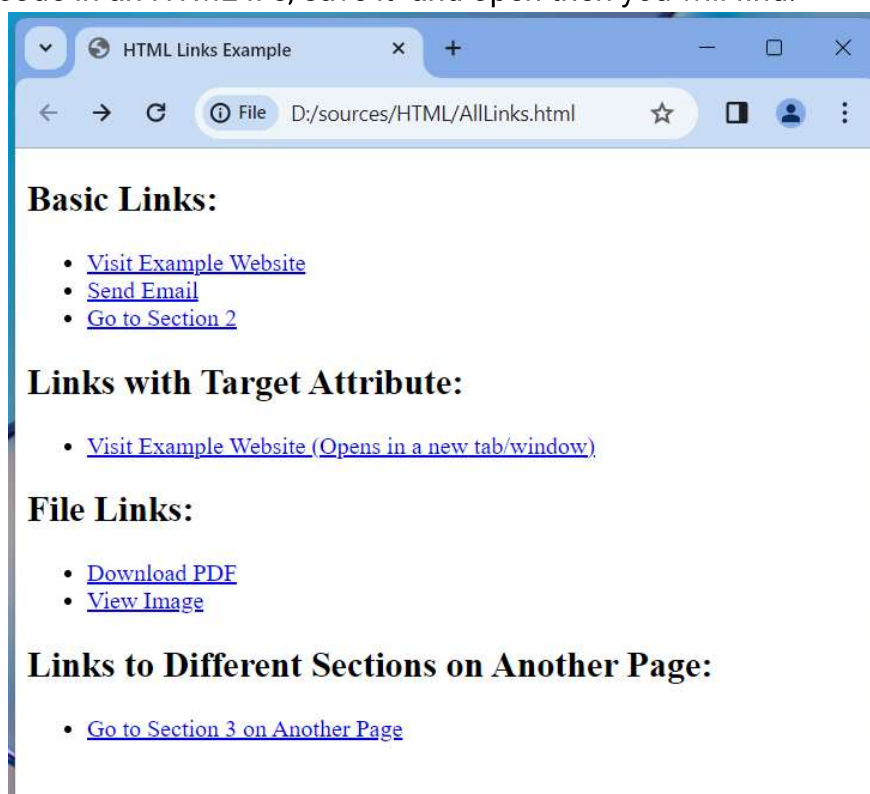
```

        <li><a href="document.pdf">Download PDF</a></li>
        <li><a href="image.png">View Image</a></li>
    </ul>

    <h2>Links to Different Sections on Another Page:</h2>
    <ul>
        <li><a href="another-page.html#section3">Go to Section 3 on Another
Page</a></li>
    </ul>
</body>
</html>

```

If you copy this code in an HTML file, save it and open then you will find:



VII. Images

In HTML, images are displayed using the `<img>` (image) element. Images can be linked from external sources, such as other Websites, or from internal sources, like files within the same directory as the HTML file. Here are some common attributes used with the `<img>` element, along with examples:

1. src (Source) Attribute:

The `src` attribute specifies the URL or file path of the image to be displayed. It's a required attribute.

Example (External Image):

```

```

In this example, the src attribute points to an external image. The alt attribute provides alternative text for the image, which is displayed if the image fails to load.

Example (Internal Image):

```

```

Here, the src attribute specifies a local image file within the same directory as the HTML file.

2. alt (Alternative Text) Attribute:

The alt attribute provides alternative text for the image. It is displayed if the image cannot be loaded and is also used by screen readers for accessibility.

Example:

```

```

3. width and height Attributes:

The width and height attributes specify the width and height of the image in pixels, respectively. These attributes are optional but recommended for controlling the layout of the page.

Example:

```

```

In this example, the image will be displayed with a width of 300 pixels and a height of 200 pixels.

Now let us create an HTML file with an image link. The HTML code is given below.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Image Link Example</title>
  <style>
    body {
      font-family: Arial, sans-serif;
```

```

        margin: 20px;
    }
</style>
</head>
<body>

    <h1>Some popular shopping portals</h1>

    <!-- Image Link -->
    <a href="https://www.amazon.com" target="_blank">

    </a>

    <a href="https://www.myntra.com" target="_blank">

    </a>

    <a href="https://www.flipkart.com" target="_blank">

    </a>

    <p>Click the image to go to the Website of the shopping portal</p>

</body>
</html>

```

Output:

Some popular shopping portals



Click the image to go to the Website of the shopping portal

4. title Attribute:

The title attribute provides additional information about the image, which is displayed as a tooltip when the user hovers over the image.

Example:

```

```

In addition to above we can also create Responsive Images with CSS (Cascading Style Sheets) which will be discussed later.

These are some of the commonly used attributes with the element in HTML. Images are crucial for enhancing the visual appeal of Websites, and understanding these attributes helps in effective image presentation and accessibility.

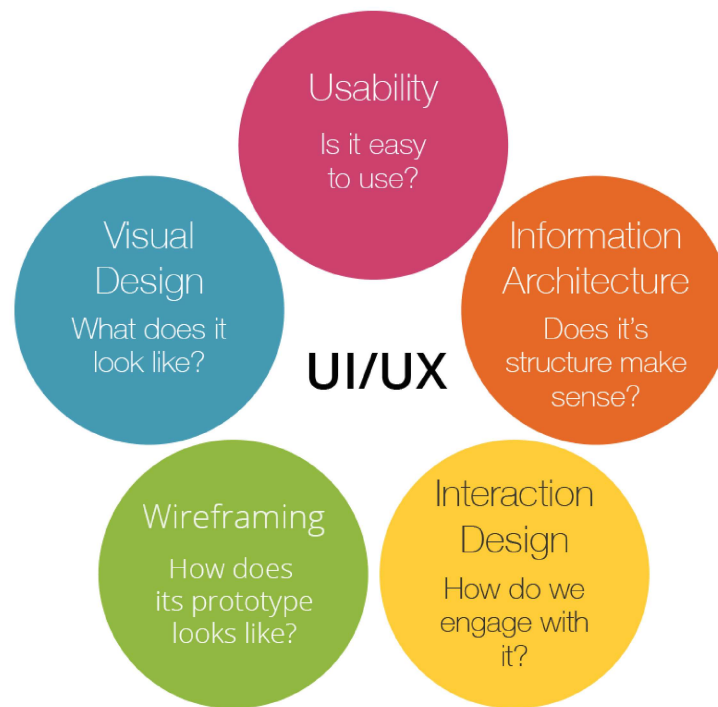
VIII. Introduction to UI/UX

UI/UX stands for User Interface (UI) and User Experience (UX), which are two critical components of the design and development process for Websites, applications, and other digital products. UI/UX Design is the process of designing the interface, optimizing navigation, and showing relevant features of a product or service.

1. **User Interface (UI):** UI refers to the visual elements and interactive components of a product that users interact with. It includes everything from buttons, icons, and forms to the overall layout and design aesthetics.
 - **Focus Areas:**
 - Visual Design: The overall look and feel of the user interface, including color schemes, typography, and imagery.
 - Interactive Design: How users interact with the product, including navigation, buttons, and other user controls.
 - **Goal:** The goal of UI design is to create an aesthetically pleasing and visually intuitive interface that is easy for users to understand and interact with.
2. **User Experience (UX):** UX encompasses the overall experience users have while interacting with a product. It goes beyond just the visual elements and includes factors like usability, accessibility, and the emotional impact of the user journey.
 - **Focus Areas:**
 - Usability: How easy and efficient it is for users to achieve their goals within the product.
 - Accessibility: Ensuring that the product is usable by people of all abilities and disabilities.
 - User Journey: The entire experience a user has, from the first interaction to completion of tasks and beyond.

- **Goal:** The goal of UX design is to create a positive and meaningful experience for users, addressing their needs and providing value.

Following diagram explains the core design criteria of UI/UX and it is important to analyze these questions while designing an attractive responsive Web Application.



Key Points:

- UI and UX are closely related but represent different aspects of the design process.
- Effective UI/UX design considers both the visual appeal (UI) and the overall user experience (UX).
- Collaboration between UI designers, UX designers, developers, and other stakeholders is crucial to creating successful digital products.
- User-centered design principles, usability testing, and continuous iteration are integral to achieving optimal UI/UX.

Companies today enjoy many advantages by adding UI (user interface) design to their development processes. These benefits may include:

- Stronger insights from user engagement
- Reduced troubleshooting and associated costs
- Improved conversion rates and reduced abandonment

- Increased customer acquisition and loyalty
- Improved search engine optimization (SEO)

There are also many advantages for companies to incorporate UX into a product. These benefits may include:

- Improved sales and revenue performance
- Reduced training and support costs
- Decreased development time and costs
- Reduced maintenance time and costs
- Improved customer engagement and satisfaction

In summary, while UI focuses on the look and feel of the interface, UX is concerned with the holistic user experience, ensuring that the product is not only visually appealing but also functional, usable, and enjoyable for its intended audience.

IX. Web Page design guidelines and best practices

Web designing is the process of planning, conceptualizing, and implementing the plan for designing a Website in a way that is functional and offers a good user experience. Web design best practices encompass a set of guidelines and principles that aim to create effective, user-friendly, and visually appealing Websites.

There are over 1.74 billion Websites on the internet today and over 4 million mobile apps available for download. So with such a huge number of options available to the customer it is very important for the companies to build a visually pleasing and efficient Website to remain competitive.

These practices are based on a combination of design, usability, and user experience principles. Here are some key Web design best practices:

Responsive Design: Ensure that your Website is designed to be responsive, meaning it adapts to different screen sizes and devices (desktops, tablets, smartphones). This improves the user experience and is essential for reaching a broader audience. With the increasing variety of devices and screen sizes, responsive design ensures a consistent and accessible experience for all users.

Multi-platform Design: Start the design process with a focus on mobile devices and progressively enhance for larger screens. This approach ensures that the Website is optimized for smaller screens first. As mobile usage continues to rise, designing for mobile first helps prioritize essential content and functionality.

Clear Navigation: Design an intuitive and easy-to-understand navigation system. Use clear menu structures, labels, and calls-to-action to guide users through your Website.

Clear navigation enhances user experience, helps visitors find information quickly, and reduces bounce rates.

Consistent Branding: Maintain consistency in branding elements such as colors, fonts, and logos throughout the Website. Consistent branding builds trust and strengthens brand identity. Brand consistency reinforces the identity of your Website and promotes a cohesive user experience.

Fast Page Load Times: Optimize images, minimize code, and leverage browser caching to ensure fast loading times. Slow-loading pages can lead to a poor user experience and negatively impact SEO. Users expect Websites to load quickly, and search engines prioritize fast-loading pages in search results.

Accessible Design: Design with accessibility in mind to ensure that people with disabilities can access and use your Website. Follow Web content accessibility guidelines (WCAG) to create a more inclusive user experience. Accessibility improves the usability of your Website and expands your audience to include individuals with diverse abilities.

Content Hierarchy and Readability: Use a clear content hierarchy with proper headings, subheadings, and formatting. Ensure text is legible with a readable font size and appropriate contrast. A well-organized content hierarchy improves readability and helps users scan and understand the information presented.

Whitespace Utilization: Use whitespace strategically to create visual clarity and prevent a cluttered appearance. Whitespace helps guide the user's focus and improves overall readability. Proper use of whitespace enhances the visual appeal of your Website and contributes to a more enjoyable user experience.

Browser Compatibility: Test your Website across different browsers to ensure consistent functionality and appearance. Address any compatibility issues that may arise. Users access Websites from various browsers, and ensuring compatibility enhances the reach and usability of your site.

Selecting Typography: Choosing the right typography is crucial for a visually appealing and readable Website. Select fonts that align with your brand identity and are easy to read across different devices. Use a combination of fonts for headings and body text that provides a clear hierarchy.

Prioritize readability by selecting fonts with clear letterforms and appropriate spacing. Use a consistent typography style across your Website for a cohesive and professional look. Ensure that the chosen fonts are Web-safe and render well on various devices and screen sizes.

Choose a Color Scheme: A well-thought-out color scheme enhances the visual appeal of your Website and reinforces your brand identity. Choose colors that complement each other and evoke the desired emotions.

Incorporate your brand colors into the design to maintain consistency with your overall brand identity. Ensure sufficient contrast between text and background colors for readability. Consider color choices that are accessible to users with different abilities, ensuring readability and usability.

Proper CTA (Call to Action): Web Design Good Practice: CTAs are critical for guiding users to take specific actions on your Website. Design them to stand out, use compelling language, and place them strategically on your pages.

Make sure the CTA's purpose is clear, and users understand the action they are expected to take. Use contrasting colors to make CTAs visually distinct from the surrounding content. Position CTAs where users are likely to notice them, such as at the end of content sections or near important information.

Limit the Options: Web Design Good Practice: Providing too many options can overwhelm users and lead to decision fatigue. Limit choices to streamline the user experience and guide users toward their goals.

- Keep the design clean and uncluttered by focusing on essential elements and minimizing unnecessary choices.
- Design a clear and intuitive user flow that guides users through the key actions and prevents confusion.
- Prioritize content and features based on user needs and business goals, offering a more focused and effective experience.

Regular Testing and Iteration: Continuously test your Website for usability, performance, and compatibility. Gather user feedback and make iterative improvements to enhance the overall user experience. Regular testing and iteration help keep your Website up-to-date, addressing evolving user needs and technological changes.

By adhering to these Web design best practices, you can create Websites that not only look visually appealing but also provide a positive and effective user experience for a diverse audience.

Note: A complete discussion on best practice Web page design is beyond the scope of this session. Here, we have just highlighted some of the major points. While understanding the theoretical principles and best practices is essential, practical experience allows designers to refine their skills and develop an intuition for effective design. Mastering the technique of creating good Web Design also needs years of experience but a Web Designer must understand the client requirements, and manage their expectations from the start of the career. A Web designer should focus on good design, get user feedback, and gain insights into user behavior. This understanding helps in making informed design decisions that cater to the preferences and expectations of the target audience.

Here are some links to understand best practices in Web design.

<https://www.hotjar.com/web-design/best-practices/>

<https://addmonte.co.uk/tips/webdesign-mistakes>

<https://www.webpartner.co.za/blog/good-vs-bad-website-design>

<https://blog.hubspot.com/blog/tabid/6307/bid/30557/6-guidelines-for-exceptional-website-design-and-usability.aspx>

<https://support.google.com/businesssitebuilder/answer/1657560?hl=en>

ChatGPT Exercise:

Please visit the site - <https://chat.openai.com/> and create your account if not yet created. Next, start with the first question on the prompt:

I am learning HTML. How do I insert special characters, copyright symbols and trademark symbols on a browser page?

I am learning HTML. How do I insert special characters, copyright symbols and trademark symbols on a browser page?



Submit the query clicking the green arrow icon. Now go through the response received from the AI engine. Output will be something like the following screenshot.



To insert special characters like the copyright symbol (©) or trademark symbol (™) into an HTML document, you can use HTML entities or numeric character references. Here's how you can do it:

Using HTML Entities:

HTML entities are sequences of characters that represent special characters. Here are the entities for the copyright and trademark symbols:

- Copyright Symbol (©): `©`
- Trademark Symbol (™): `™`

To display the copyright symbol, for example, you would use `©` in your HTML code. Here's an example:

```
html
<p>&copy; 2023 Your Company Name. All rights reserved.</p>
<p>Product Name&trade; is a registered trademark.</p>
```

Copy code

Regenerate

Send a message

You can also get help on all symbols and all special characters by changing the question. Ask the question “Please help me display all the special characters in HTML?”

After the 2 hr ILT, the student has to do a 1 hour live lab. Please refer to the Day 2 Lab doc in the LMS.